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23K-0004

Agentic AI

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Assignment # 02

Q1 23K-0004

R = 04 , D = 2+3+0+0+0+4 => 9

Parameter	Formula
Normal daily queries, Q	1500 + 25×4 => 1600
LLM calls per query, N	5+(4 mod 4) => 5
Tool call failure prob per call, p	0.04 + 0.01×9 => 0.13
Heatwaves per month, H	4+(4 mod 4) => 4
Monthly inference budget, B	1200 + 10×4 => 1240$
End-to-end latency target, T	10 - (4 mod 3) => 9

Token Growth: input token = 2000 + 800(k-1)

Section 2:

1. For 5 calls:

Call	Input	Output
1	2000+800(0) => 2000	350
2	2000+800(1) => 2800	350
3	2000+800(2) => 3600	350
4	2000+800(3) => 4400	350
5	2000+800(4) => 5200	600

Base input = 2000+2800+3600+4400+5200 => 18000

Base output = (4×350) +600 => 2000

Expected Input Retries = 0.13 (2000+2800+3600+4400) => 1664

Expected Output Retries = 0.13×350×4 => 182

Expected input tokens = 18000 + 1664 => 19664

Expected output tokens = 2000 + 182 => 2182

Normal days = 30 - 4 => 26

Q_{normal} = 26 (1600) => 41,600

$Q_{heatwave}$ = 4 (5×1600) => 32,000

Monthly queries = 41600 + 32,000 => 73,600

Monthly input tokens = 73,600 × 19664 => 1,447,270,400

Monthly output tokens = 73,600 × 2182 => 160,595,200

Total = 1,607,865,600

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2). Input = 1447270400 / 1000000 => 1447.2704 million

Output = 160595200 / 1000000 => 160.5952 million

Expected LLM calls = 5 + (0.13 × 4) => 5.52

cached prefix tokens = 5.52 × 1500 => 8280

monthly cacheable prefix tokens = 8280 × 73600 => 609.408 million

Remaining input tokens = 1447.2704 - 609.408 => 837.8624 million

assume: initial cache write = 1500 tokens

73600 × 1500 => 110.4 million cache writes

609.408 - 110.4 => 499.008 million cache reads

No cache Cost = 1447.2704 (P_i) + 160.5952 (P_o)

Ideal cache Cost = 837.8624 (P_i) + 160.5952 (P_o) + 609.408 (P_c)

Cache with write: 837.8624(P_i) + 160.5952(P_o) + 110.4 (P_w) + 499.008 (P_c)

Model	No cache	Ideal Cache	Read + write cache
Claude Opus 5.5	$9000.9856	$6685.2352	$7215.1552
Claude Sonnet 5.5	$4500.4928	$3342.6176	$3607.5776
Gemini 4 Argon	$4500.4928	$3342.6176	$3342.6176
GPT-6 Astra	$22502.464	$17017.792	$18287.392
Grok 4.7	$3858.112	$2944	$3164.8
GPT-6 Sol	$4500.4928	$3403.5584	$3657.4784
gpt-oss-120b	$70.7193	$70.7193	$70.7193
Llama 4 Scout	$192.9056	$192.9056	$192.9056
Kimi K3	$6750.7392	$5105.3376	$5403.4176
Deepseek V4 Flash 1M	$313.4477	$223.8647	$240.0935
Nemotron 3.5 super VL	$334.3633	$242.9521	$263.9281
Qwen 3.5-9B	$136.659	$112.2827	$116.6987

3). Heatwave daily queries = 5 × 1600 => 8000

Peak queries = 0.35 × 8000 => 2800

2 peak hrs => 120 mins

Query per min = 2800/120 => 23.33

RPM = 23.33 × 5.52 => 128.8

Input TPM = 23.33 × 19664 => 458826.6667

Output TPM = 23.33 × 2182 => 50913.33

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Q2.

Section 1: $r = 16$, $W_0 \rightarrow 4096 \times 4096$

a). $P_{LoRA} = r(d_{in} + d_{out}) \Rightarrow 16(4096 + 4096) \Rightarrow 131072$

b). $P_{original} = 4096 \times 4096 \Rightarrow 16777216$

$$\frac{131072}{16777216} \times 100 \Rightarrow 0.78125\%.$$

c). Initializing $B = 0$ and A randomly preserves the pretrained model's behavior at the beginning while allowing the adapter to learn. Randomly initializing both matrices could introduce unnecessary disturbances to the pretrained weights.

d). QLoRA: LoRA reduces trainable parameters but the frozen base model is still stored in FP16 or BF16 requiring significant GPU memory. QLoRA uses 4-bit Normal Float (NF4) quantization to store the pretrained model's weights more efficiently. It uses also paged optimizers to help manage temporary GPU memory spikes.

DoRA: Standard LoRA uses low-rank update to modify weights, which may limit its ability to reproduce the flexibility of full fine-tuning. DoRA separates the weight matrix into magnitude and direction components to provide additional flexibility.

AdaLoRA: Standard LoRA uses a fixed rank for every selected layer, while AdaLoRA dynamically allocates more rank capacity to important weight matrices and reduces capacity in less important matrices.

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Q3. $y = [1, 0, 0, 0]$

$z_t = [5.0, 3.0, 2.0, 0.0]$

$z_s = [3.0, 2.5, 1.0, 0.5]$

$T = 2, \quad \alpha = 0.5 \quad \eta = 0.5$

a) Teacher: $p_{11} = \frac{e^5}{e^5+e^3+e^2+e^0}$, $p_{12} = \frac{e^3}{e^5+e^3+e^2+e^0}$, $p_{13} = \frac{e^2}{e^5+e^3+e^2+e^0}$, $p_{14} = \frac{e^0}{e^5+e^3+e^2+e^0}$

Teacher Prob = $[0.839, 0.1135, 0.04178, 0.00565]$

Student: $q_{11} = \frac{e^3}{e^3+e^{2.5}+e^1+e^{0.5}}$, $q_{12} = \frac{e^{2.5}}{e^3+e^{2.5}+e^1+e^{0.5}}$, $q_{13} = \frac{e^1}{e^3+e^{2.5}+e^1+e^{0.5}}$, $q_{14} = \frac{e^{0.5}}{e^3+e^{2.5}+e^1+e^{0.5}}$

Student Prob = $[0.5483, 0.3325, 0.0742, 0.045]$

b) $z_T = [5, 3, 2, 0]/2 \Rightarrow [2.5, 1.5, 1, 0]$

Teacher: $p_{T1} = \frac{e^{2.5}}{e^{2.5}+e^{1.5}+e+1}$, $p_{T2} = \frac{e^{1.5}}{e^{2.5}+e^{1.5}+e+1}$, $p_{T3} = \frac{e}{e^{2.5}+e^{1.5}+e+1}$, $p_{T4} = \frac{1}{e^{2.5}+e^{1.5}+e+1}$

Prob = $[0.5977, 0.2199, 0.1334, 0.04906]$

Student: $z_s = [3, 2.5, 1, 0.5]/2 \Rightarrow [1.5, 1.25, 0.5, 0.25]$

$q_{T1} = \frac{e^{1.5}}{e^{1.5}+e^{1.25}+e^{0.5}+e^{0.25}}$, $q_{T2} = \frac{e^{1.25}}{e^{1.5}+e^{1.25}+e^{0.5}+e^{0.25}}$, $q_{T3} = \frac{e^{0.5}}{e^{1.5}+e^{1.25}+e^{0.5}+e^{0.25}}$, $q_{T4} = \frac{e^{0.25}}{e^{1.5}+e^{1.25}+e^{0.5}+e^{0.25}}$

Prob = $[0.411, 0.3201, 0.1512, 0.1177]$

c) $KL(p_2 \| q_2) = \sum_i p_{2i} \ln(p_{2i}/q_{2i})$

"Hospital": $0.5977 \ln(0.5977/0.411) \Rightarrow 0.2238$

"Clinic": $0.2199 \ln(0.2199/0.3201) \Rightarrow -0.08256$

"Ward": $0.1334 \ln(0.1334/0.1512) \Rightarrow -0.01671$

"School": $0.04906 \ln(0.04906/0.1177) \Rightarrow -0.04293$

$KL = 0.2238 - 0.08256 - 0.01671 - 0.04293 \Rightarrow 0.0816$

d) $CE(y, q_1) = -\sum_i y_i \ln(q_{1i})$

$y = [1, 0, 0, 0]$

$CE = -\ln(0.5483) \Rightarrow 0.6009$

e) $Loss = 0.5(2)^2(0.0816) + (1-0.5)(0.6009)$

$Loss = 0.46365$

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f). $\frac{\partial L}{\partial z_{s,i}} = \alpha T(q_{T,i} - p_{T,i}) + (1-\alpha)(q_{1,i} - y_i)$

$G_1 = (0.411 - 0.5977) + 0.5(0.5483 - 1) \Rightarrow -0.41255$

$G_2 = (0.3201 - 0.2199) + 0.5(0.3325 - 0) \Rightarrow 0.26645$

$G_3 = (0.1512 - 0.1334) + 0.5(0.0742) \Rightarrow 0.0549$

$G_4 = (0.1177 - 0.04906) + 0.5(0.045) \Rightarrow 0.09114$

$G = [-0.41255, 0.26645, 0.0549, 0.09114]$

g). $z_{new,i} = z_{old,i} - \eta(\partial L / \partial z_{s,i})$

$z_{new,1} = 3 - 0.5(-0.41255) \Rightarrow 3.2063$

$z_{new,2} = 2.5 - 0.5(0.26645) \Rightarrow 2.3668$

$z_{new,3} = 1 - 0.5(0.0549) \Rightarrow 0.9726$

$z_{new,4} = 0.5 - 0.5(0.09114) \Rightarrow 0.4544$

$z_s = [3.2063, 2.3668, 0.9726, 0.4544]$

$T = 1$: $e^{3.2063} + e^{2.3668} + e^{0.9726} + e^{0.4544} \Rightarrow 39.5708$

$q_{1,new} = \left[\frac{24.6876}{39.5708}, \frac{10.6632}{39.5708}, \frac{2.6448}{39.5708}, \frac{1.5752}{39.5708}\right]$

$\Rightarrow [0.6239, 0.2695, 0.06684, 0.03981]$

$T = 2$: $z_s = [1.6032, 1.1834, 0.4863, 0.2272]$

$e^{1.6032} + e^{1.1834} + e^{0.4863} + e^{0.2272} \Rightarrow 11.1157$

$q_{T,new} = \left[\frac{4.9689}{11.1157}, \frac{3.2654}{11.1157}, \frac{1.5263}{11.1157}, \frac{1.2551}{11.1157}\right]$

$\Rightarrow [0.447, 0.2938, 0.1463, 0.1129]$

KL:

"Hospital": $0.5977 \ln(0.5977/0.447) \Rightarrow 0.1736$

"Clinic": $0.2199 \ln(0.2199/0.2938) \Rightarrow -0.06371$

"Ward": $0.1334 \ln(0.1334/0.1463) \Rightarrow -0.01231$

~~"School": $0.1177 \ln(0.1177/0.1129) \Rightarrow$~~

"School": $0.04906 \ln(0.04906/0.1129) \Rightarrow -0.04089$

$KL = 0.1736 - 0.06371 - 0.01231 - 0.04089$

$KL = 0.05669$

CE: $-\ln(0.6239) \Rightarrow +0.4718$

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Loss = $(0.5)(2^2)(0.05669) + 0.5(0.4718)$

Loss = 0.3493

Q4.

1. Acc = $\frac{100+80+50}{300} \Rightarrow 0.7667 \Rightarrow 76.67\%$

Wilson score method for CI: $z = 1.96$

center: $\frac{\hat{p} + (z^2/2n)}{1 + (z^2/n)}$, half-width: $z\frac{\sqrt{\hat{p}(1-\hat{p})/n + (z^2/4n^2)}}{(1 + z^2/n)}$

center = $\frac{0.7667 + (1.96^2/600)}{1 + (1.96^2/300)} \Rightarrow 0.7633$

half-width = $\frac{1.96\sqrt{(0.7667(1-0.7667)/300) + (1.96^2/4\times300^2)}}{1 + (1.96^2/300)}$

$\Rightarrow \frac{1.96 \times 0.02464}{1.0128} \Rightarrow 0.04768$

95% Wilson CI = 0.7633 ± 0.04768

$\Rightarrow [0.71562, 0.81098]$

2. Positive:

TP = 100, FP = 20+8 => 28, FN = 5+15 => 20

Precision = $\frac{100}{100+28} \Rightarrow 0.78125$, Recall = $\frac{100}{100+20} \Rightarrow 0.8333$, F1 = $\frac{2P.R}{P+R}$

$\Rightarrow 0.8064$

Negative:

TP = 80, FP = 5+10 => 15, FN = 8+12 => 20

Precision = $\frac{80}{80+15} \Rightarrow 0.8421$, Recall = $\frac{80}{80+20} \Rightarrow 0.8$, F1 = $\frac{2\times0.8421\times0.8}{0.8421+0.8} \Rightarrow 0.8205$

Neutral:

TP = 50, FP = 15+12 => 27, FN = 20+10 => 30

Precision = $\frac{50}{50+27} \Rightarrow 0.6494$, Recall = $\frac{50}{50+30} \Rightarrow 0.625$, F1 = $\frac{2\times0.6494\times0.625}{0.6494+0.625} \Rightarrow 0.637$

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3. Macro-F1:

$\Rightarrow \frac{0.8064+0.8205+0.637}{3} \Rightarrow 0.7546 \Rightarrow 75.46\%$

Weighted F1:

$\Rightarrow \frac{0.8064(120)+0.8205(100)+0.637(80)}{300} \Rightarrow 0.7659 \Rightarrow 76.59\%$

Micro F1:

$\Rightarrow$ TP = 230, FP = 70

Micro Precision: $230/300 \Rightarrow 0.7667$

Micro Recall: $230/300 \Rightarrow 0.7667$

Micro F1 = 0.7667

4. The LLM struggles the most in "Neutral Class" as shown by evaluation metrics. & It is strongly recommended to include a few examples of neutral classes (Few-shot) for better classification of mixed examples.

5. Acc: $\frac{106+84+64}{300} \Rightarrow 0.8467 \Rightarrow 84.67\%$

$\text{center} = \frac{0.8467 + (1.96^2/600)}{1+(1.96^2/300)} \Rightarrow 0.8423$

$\text{half-width} = \frac{1.96\sqrt{[0.8467(1-0.8467))/300 + (1.96^2/4\times300^2)}}{1 + 1.96^2/300}$

$\Rightarrow \frac{1.96 \times 0.0211}{1.0128} \Rightarrow 0.04083$

95% Wilson CI = 0.8423 ± 0.04083

$\Rightarrow [0.80147, 0.88313]$

Positive class:

TP = 106, FP = 6+9 $\Rightarrow$ 15, FN = 4+10 $\Rightarrow$ 14

Precision = $\frac{106}{106+15} \Rightarrow 0.876$, Recall = $\frac{106}{106+14} \Rightarrow 0.8833$, F1 = $\frac{2\times0.876\times0.8833}{0.876+0.8833}$

$\Rightarrow 0.8796$

Negative class:

Precision = TP = 84, FP = 4+7 $\Rightarrow$ 11, FN = 6+10 $\Rightarrow$ 16

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Precision = $\frac{84}{84+11}$ $\Rightarrow 0.8842$, Recall = $\frac{84}{84+16}$ $\Rightarrow 0.84$, F1 = $\frac{2 \cdot 0.8842 \times 0.84}{0.8842 + 0.84} \Rightarrow 0.8615$

Neutral class:

TP = 64, FP = 10+10 $\Rightarrow$ 20, FN = 9+7 $\Rightarrow$ 16

Precision = $\frac{64}{64+20}$ $\Rightarrow 0.7619$, Recall = $\frac{64}{64+16}$ $\Rightarrow 0.8$, F1 = $\frac{2 \times 0.7619 \times 0.8}{0.7619 + 0.8} \Rightarrow 0.7805$

Macro-F1:

$\Rightarrow \frac{0.8796 + 0.8615 + 0.7805}{3} \Rightarrow 0.8405 \Rightarrow 84.05\%$

Weighted F1:

$\Rightarrow \frac{0.8796(120) + 0.8615(100) + 0.7805(80)}{300} \Rightarrow 0.84714 \Rightarrow 84.714\%$

Micro-F1:

TP = 254, FN = 46

Micro-F1 = 0.8467 $\Rightarrow$ 84.67%

Comparison with Stage 1:

Metric	Stage 1	Stage 2
Accuracy	0.7667	0.8467
Macro F1	0.7546	0.8405
Weighted F1	0.7659	0.84714
Micro F1	0.7667	0.8467
"Neutral" Recall	0.625	0.8

6. Acc = [0.847, 0.84, 0.853, 0.837, 0.85]

$\bar{x} = \frac{0.847 + 0.84 + 0.853 + 0.837 + 0.85}{5} \Rightarrow 0.8454$

$\sigma^2 = \frac{1}{5-1}[(0.847-0.8454)^2 + (0.84-0.8454)^2 + (0.853-0.8454)^2 + (0.837-0.8454)^2 + (0.85-0.8454)^2]$

$\sigma^2 = \frac{1}{4}(2.56\times10^{-6} + 2.916\times10^{-5} + 5.776\times10^{-5} + 7.056\times10^{-5} + 2.116\times10^{-5})$

$\sigma^2 = \frac{1}{4}(1.812\times10^{-4}) \Rightarrow 4.53\times10^{-5}$

$\sigma = \sqrt{4.53\times10^{-5}} \Rightarrow 0.006705$

Mean $\pm \sigma \Rightarrow 0.8454 \pm 0.006705$

$[0.8387, 0.8521]$

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8. The prompt is highly stable but nonetheless, different orderings impact the result slightly

7. Reference: 10 words

Candidate: 9 words

BLEU-4:

Unigram: 8/9 candidate words match with reference.

Bigrams: 5/8 candidate match with reference

the product, product quality, quality is, is good, delivery was

Trigrams: 3/7

the product quality, product quality is, quality is good

4-grams: 2/6

the product quality is, product quality is good

$B.P = e^{1-10/9} \Rightarrow 0.8948$

$BLE-4 = 0.8948 \times (0.8889 \times 0.625 \times 0.4286 \times 0.3333)^{1/4}$

$\Rightarrow 0.4749$

Rouge-1:

matching unigrams: 8/9, $c = 9$, $r = 10$

Rouge-1 Precision $= \frac{8}{9} \Rightarrow 0.8889$, Rouge-1 Recall $= \frac{8}{10} \Rightarrow 0.8$

Rouge-1 F1 $= \frac{2 \times 0.8889 \times 0.8}{0.8889 + 0.8} \Rightarrow 0.8421$

Rouge-2:

matching bigrams $= 5$, $r = 10-1 \Rightarrow 9$, $c = 9-1 \Rightarrow 8$

Rouge-2 Precision $= \frac{5}{8} \Rightarrow 0.625$, Rouge-2 Recall $= \frac{5}{9} \Rightarrow 0.5555$

Rouge-2 F1 $= \frac{2 \times 0.625 \times 0.5555}{0.625 + 0.5555} \Rightarrow 0.5882$

Rouge-L: LCS: "the product quality is good" $l = 5$

Rouge-L Precision $= \frac{5}{9} \Rightarrow 0.5555$, Rouge-L Recall $= \frac{5}{10} \Rightarrow 0.5$

Rouge-L F1: $\frac{2 \times 0.5555 \times 0.5}{0.5555 + 0.5} \Rightarrow 0.5263$

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METEOR:

matching chunks: "the product quality is good", "but", "delivery was" => 3

matches = 8, c = 9, r = 10

Precision = $\frac{8}{9}$ => 0.8889, Recall = $\frac{8}{10}$ => 0.8

F-mean = $\frac{10 \times 0.8889 \times 0.8}{0.8 + (9 \times 0.8889)}$ => 0.80808

Fragmentation Penalty = $0.5 \times (3/8)^3$ => 0.02637

METEOR = $0.80808 \times (1 - 0.02637)$ => 0.7868

8. [0.6, 0.5, 0.4, 0.8, 0.7, 0.3, 0.25, 0.5, 0.2]

Perplexity = $e^{-1/N \sum \ln(p_i)}$

$\sum \ln(p_i) = \ln(0.6) + \ln(0.5) + \ln(0.4) + \ln(0.8) + \ln(0.7) + \ln(0.3) + \ln(0.25) + \ln(0.5) + \ln(0.2)$

=> -7.5929

Perplexity = $e^{+9.5929/9}$ => 2.3248

This means that the sequence has this average log-loss ~~equivalent~~ Σ choices per step. Low perplexity signals higher model confidence in its generated tokens, not factual correctness.